SAMARTH

CUET PG 2027

Lecture – 17

General LLB

Unit 3 : Digital Fundamental

Topic : Previous Year Questions + Doubt Class

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Recap *of previous lecture*

1. Memory Unit



Topics *to be covered*

1. Previous Year Questions + Doubt Class



Topic : Previous Year Questions + Doubt Class

Topic : Previous Year Questions + Doubt Class

#Q. Given that numbers A and B are two 8 bit 2's Complement numbers with A = 11111111; B = 11111111. Then sum A + B is _____

(A) 00000010

(B) 11111100

(C) 11111110

(D) 00000000



Ans. C

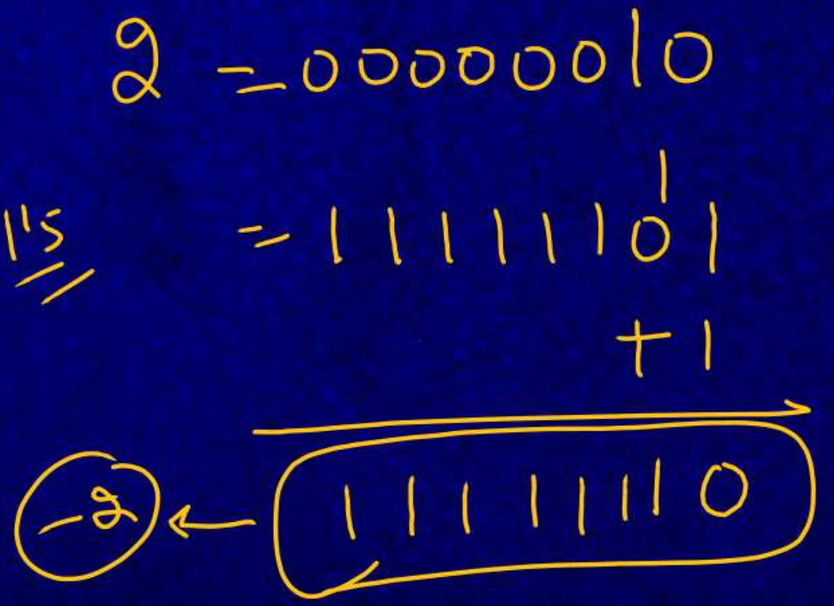
Solution:

Convert A and B to decimal

- A = 11111111 in 2's complement is -1.
- B = 11111111 in 2's complement is -1.

Sum A + B :

A+B = -1+(-1) = -2



Now, we need to represent -2 in 8-bit 2's complement.

So, -2 = 11111110 in 8-bit 2's complement.

Correct Answer: Option (C) 11111110

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SOP ⇒ consider O/P 1

#Q. The Boolean expression for the following truth table is _____

x	y	z	f
0	0	0	0
0	0	1	0
0 $\bar{x}$	1 y	0 $\bar{z}$	1
0	1	1	0
1	0	0	0
1 x	0 $\bar{y}$	1 z	1
1	1	0	0
1 x	1 y	1 z	1

$\bar{x}y\bar{z}$

$x\bar{y}z$

xyz

Input

0 → Complement

1 → without complement

$\bar{x}yz + x\bar{y}z + xyz$



(A) F = x'yz' + xy'z + x'y'z'

(B) F = x'y'z' + xy'z + xyz'

(C) F= x'yz' + xy'z + xyz

(D) None of these

Ans. C
Solution:
List the minterms where f = 1
From the table:

x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	1 → minterm 2
0	1	1	0
1	0	0	0
1	0	1	1 → minterm 5
1	1	0	0
1	1	1	1 → minterm 7

So f = 1 for these input combinations:

- x=0, y=1, z=0 → x' y z'
- x=1, y=0, z=1 → x y' z
- x=1, y=1, z=1 → x y z

Boolean Expression

$f(x, y, z) = x'yz' + xy'z + xyz$

This matches: Option (C): F = x'yz' + xy'z + xyz

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#Q. Consider the following 4-bit binary numbers represented in the 2's complement form: 1101 and 0100.

What would be the result when we add them?

(A) 0001 and an overflow

(B) 1001 and no overflow

(C) 1001 and an overflow

(D) 0001 and no overflow

Paid +

YT

2's complement Arithmetic



Ans. D

Solution:

2's Complement Addition (4-bit)

Given: 1101 and 0100 (in 2's complement)

- 1101 = -3 (in decimal)
- 0100 = +4 (in decimal)
- Sum = -3 + 4 = +1
- +1 in 4-bit 2's complement = 0001

In 2's complement, overflow occurs only when:

- Both operands are positive (sign bit = 0)

→ If result has sign bit = 1 → Overflow

- Both operands are negative (sign bit = 1)

→ If result has sign bit = 0 → Overflow

In our case :

- One number is negative, one is positive → No overflow possible

Final Answer: (D) 0001 and no overflow

Topic : Previous Year Questions + Doubt Class

#Q. Which of the following is the smallest unit of data in a computer?

(A) Byte
(B) Bit
(C) Nibble
(D) KB

Bit 0,1
Nibble 4 bits
Byte 8 bit
KB. 1024 bits.



Ans. B

Solution:

- A bit (short for binary digit) is the smallest unit of data in a computer.
- It can have only two values: 0 or 1.

The correct answer is: (B) Bit

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#Q. Which of the following components is used to establish a communication link between a CPU and the peripheral devices to transfer data?

add hold

(A) Memory address register

(B) Instruction register

(C) Memory data register

(D) Index register



Ans. C

Solution:

- The Memory Data Register (MDR) is used to hold data that is being transferred to or from memory or I/O devices.
- It acts as a buffer between the CPU and the memory or I/O devices, allowing data transfer in either direction.
- In systems where memory and I/O share the same bus (memory-mapped I/O), the MDR is involved in reading/writing to I/O.

The correct answer is: (C) Memory Data Register (MDR)

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#Q. The primary purpose of cache memory in a computer system is

(A) to manage input and output operations between the CPU and peripherals

(B) to temporarily store frequently accessed data and instructions for faster access by the CPU

(C) to permanently store data and programs

(D) to provide additional storage space when the main memory is full



Ans. B

Solution:

- Cache memory is a small, fast memory located close to the CPU.
- It stores copies of frequently accessed data/instructions from main memory (RAM).
- This reduces the average time the CPU takes to access data, thus improving system performance.

The correct answer is: (B) To temporarily store frequently accessed data and instructions for faster access by the CPU

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#Q. Cache memory functions as an intermediary between

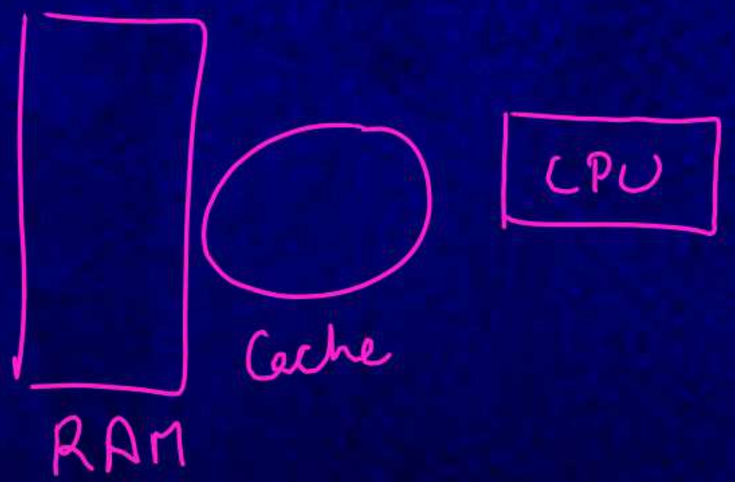
(A) RAM and ROM

(B) CPU and RAM ✓

(C) CPU and Hard Disk

(D) None of these





Ans. B

Solution:

- Cache memory is a small, high-speed memory located close to the CPU.
- It acts as an intermediary between the CPU and main memory (RAM).
- Frequently accessed data and instructions from RAM are stored in the cache so that the CPU can access them faster, reducing latency.

Correct Answer: (B) CPU and RAM

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#Q. The expression P + QR is the reduced form of _____

(A) (P + Q)R

(B) (P + R)Q

(C) (P + Q) (P + R)

(D) PQ + QR

$P + QR$

$= (P+Q)(P+R)$



Ans. C

Solution:

Option (C):

$(P+Q)(P+R) = PP+PR+PQ+QR$ (using distributive law)

$=P+PR+PQ+QR$ (since $P \cdot P=P$)

$=P+QR$ (absorption: $P+PQ=P$, $P+PR=P$)

This simplifies exactly to P + QR

Correct Answer: (C) (P + Q)(P + R)

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#Q. Match the List – 1 with List – II

List – I	List – II
(A) Asynchronous	(I) A pulse that cause a logic device to be activated or change state
(B) Trigger	(II) The operation is not executed in step with the clock
(C) J-K Flip flop	(III) Flip flop and atleast set, reset, toggle and hold modes of operation.
(D) D Flip flop	(IV) Flip flop with atleast set and rest modes of operations.

Choose the correct answer from the options given below:

A. (A)-(I),(B)-(II),(C)-(III),(D)-(IV)
B. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
C. (A)-(III),(B)-(IV),(C)-(II),(D)-(I)
D. (A)-(I),(B)-(II),(C)-(IV),(D)-(III)

Ans. B

Solution-

(A) Asynchronous

- Means not synchronized with the clock → (II) The operation is not executed in step with the clock

(B) Trigger

- A pulse that causes a logic device to change state → (I) A pulse that cause a logic device to be activated or change state

(C) J-K Flip flop

- A flip-flop with set, reset, toggle, and hold modes → (III) Flip flop with at least set, reset, toggle and hold modes of operation

(D) D Flip flop

- A flip-flop with set and reset modes only → (IV) Flip flop with at least set and reset modes of operation

Correct Option B: (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Topic : Previous Year Questions + Doubt Class

#Q. A logic circuit, that can add two 1 -bit numbers and produce output for sum and carry but cannot handle carry input, is called ..... .

(A) Half adder

(B) Full adder → Add 3 bits.

(C) Multiplexer

(D) Encoder



Ans. A

Solution:

A Half Adder is a basic logic circuit used to add two single-bit numbers.

- It has 2 inputs: A and B
- It gives 2 outputs:
- Sum
- Carry

But it cannot add carry from a previous stage — that's why it's called a half adder (it does only half the job).

In contrast, a Full Adder can:

- Add two bits and a carry input from a previous addition.

Correct answer: A. Half Adder

Topic : Previous Year Questions + Doubt Class

#Q. Match List – I with List – II.

List – I	List – II
(A) Flash Memory	(I) Oldest and Slowest
(B) PMOS	(II) Used in large scale integration (LSI)
(C) NMOS	(III) Least power consumption
(D) CMOS	(IV) Non volatile RAM which is powered continuously

Choose the correct answer from the options given below:

(A) (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
(B) (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
(C) (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
(D) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Ans. D

Solution:

(A) Flash Memory → (IV)

- Flash memory is a type of non-volatile RAM that retains data even when power is turned off. It is often used in devices like USB drives, SSDs, and memory cards.

(B) PMOS → (I)

- PMOS (P-type Metal-Oxide-Semiconductor) is one of the oldest and slowest types of MOSFET technology. It was widely used in early integrated circuits.

(C) NMOS → (II)

- NMOS (N-type Metal-Oxide-Semiconductor) is commonly used in large-scale integration (LSI) due to its faster switching speed compared to PMOS.

(D) CMOS → (III)

- CMOS (Complementary Metal-Oxide-Semiconductor) is known for its low power consumption because it uses both PMOS and NMOS transistors in a complementary way, reducing static power dissipation.

Correct option D: (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Topic : Previous Year Questions + Doubt Class

#Q. The function represented by the k-map given below is

K-map

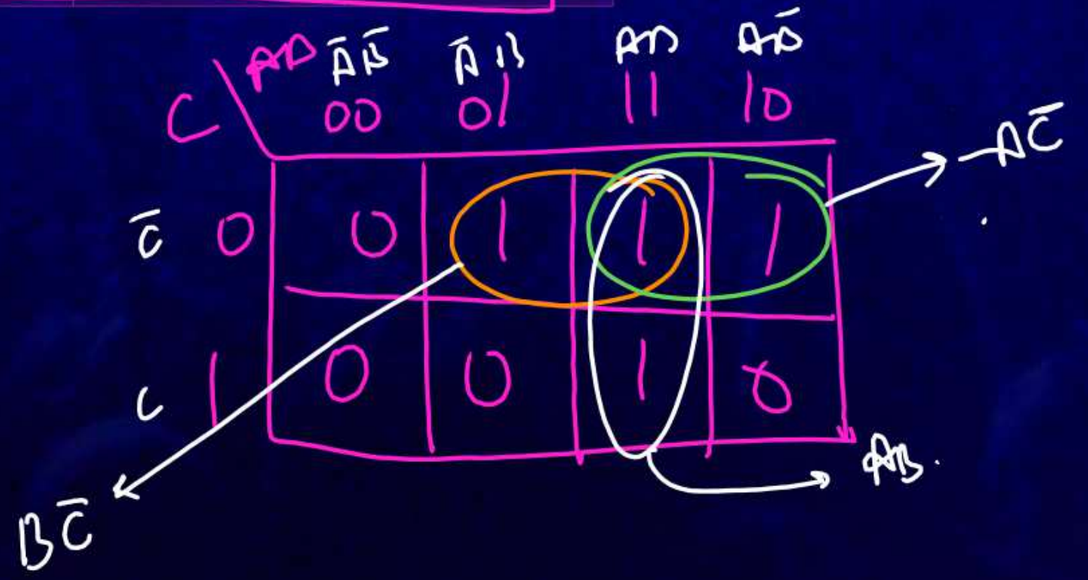
C \ AB	00	01	11	01
0	0	1	1	1
1	0	0	1	0

(A) BC+AB+AC′

(B) BC′+AB+AC′

(C) (B⊕C)′

(D) A. BC

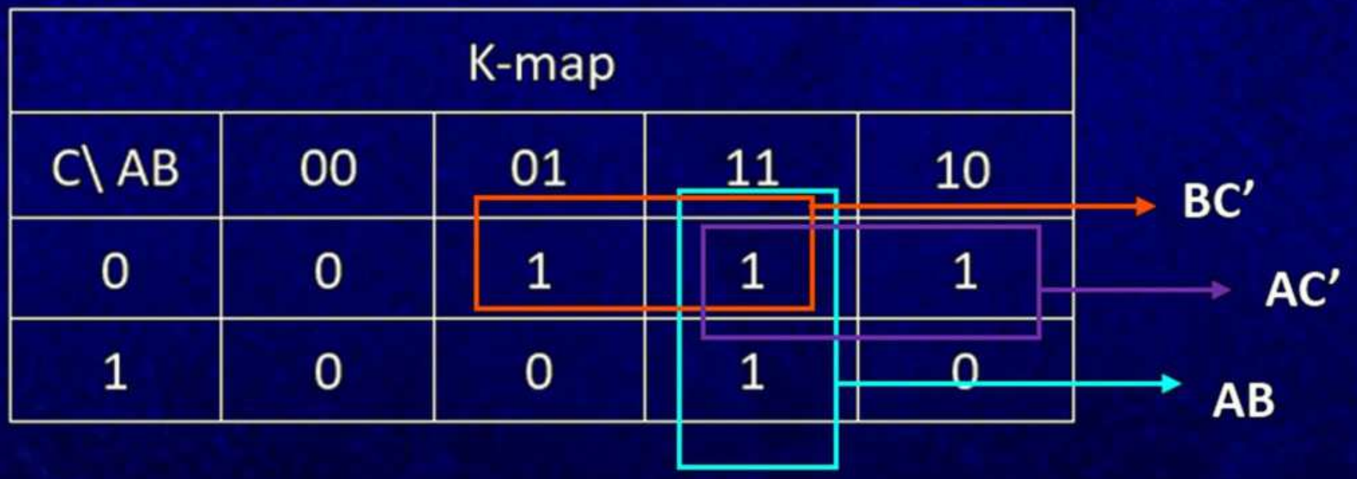


Ans. B

Solution:

- Solving K-map by grouping the 1s and then deriving the simplified Boolean expression.

K-map				
C\ AB	00	01	11	10
0	0	1	1	1
1	0	0	1	0



Correct Option B: BC′ + AB + AC′

Topic : Previous Year Questions + Doubt Class

#Q. The number of 1's in the binary representation of (3 * 4096+15 * 256+5 * 16+3) are

(A) 8

(B) 9

(C) 10

(D) 12



Ans. C

Solution:

Expression:

$3\times4096+15\times256+5\times16+3$

Evaluate each term

- $3 \times 4096 = 12288$
- $15 \times 256 = 3840$
- $5 \times 16 = 80$
- $3 = 3$

Add all values

$12288+3840+80+3=16211$

Convert 16211 to binary

We can convert 16211 to binary directly or use a calculator:

$16211_{10}=11111101100011_2$

Count the number of 1's = 10

Correct Answer: Option C — 10

Topic : Previous Year Questions + Doubt Class

#Q. Representation of -11 in sign-and magnitude is:

(A) 11011

(B) 010111

(C) 10011

(D) 01110



Ans. A

Solution:

The most significant bit (MSB) is the sign bit:

- 0 = positive
- 1 = negative

The remaining bits represent the magnitude (absolute value) in binary.

Find Binary of +11

- 1110=10112 (That's 4 bits for magnitude.)

Add Sign Bit for –11

- Sign bit = 1 (for negative)
- Magnitude = 1011

Final answer: 1 1011 = 11011 Option A.

Topic : Previous Year Questions + Doubt Class

#Q. During the execution of an instruction in a processor

(A) Program Counter (PC) is used to hold the address of next instruction to be executed.
(B) Instruction Register (IR) hold the instruction to be executed.
(C) Memory Address Register (MAR) is used to perform address translation.
(D) Data Register (DR) is used to perform operation on the data.
(E) Clock generates necessary control signals

Choose the correct answer from the options given below :

(A) Only (A) and (B) true
(B) Only (C) and (D) true
(C) Only (A), (B) and (E) true
(D) Only (B), (C) and (D) true

Ans. C

Solution:

(A) Program Counter (PC): The PC holds the memory address of the next instruction to be fetched and executed. This is crucial for the sequential execution of instructions.

(B) Instruction Register (IR): The IR stores the instruction that is currently being executed. It holds the opcode (operation code) and operands of the instruction.

(C) Memory Address Register (MAR): The MAR holds the memory address of the data to be accessed. It's used for both fetching instructions and accessing data. However, address translation (converting logical addresses to physical addresses) is primarily handled by the Memory Management Unit (MMU).

(D) Data Register (DR): The DR holds the data that is being operated on by the processor. It's used for arithmetic and logical operations.

(E) Clock: The clock generates the timing signals that control the various stages of the instruction cycle. These signals ensure that each step happens at the correct time.

Therefore, only statements (A), (B), and (E) are directly related to the execution of an instruction in a processor.

Topic : Previous Year Questions + Doubt Class

#Q. Find the sum of following hexadecimal numbers $(7CA)_{16}+(4B2)_{16}$= ?

(A) B7C

(B) C7C

(C) D7C

(D) B6C



Ans. B

Solution:

Convert both hexadecimal numbers to decimal.

$(7\mathrm{CA})_{16} = 7 \times 16^2 + 12 \times 16^1 + 10 \times 16^0$

$= 7 \times 256 + 12 \times 16 + 10$

$= 1792 + 192 + 10 = 1994 - (4\mathrm{B2})_{16} = 4 \times 256 + 11 \times 16 + 2$

$= 1024 + 176 + 2 = 1202$

Add the decimal values

$1994 + 1202 = 3196$

We convert **3196** to hex:

- $3196 \div 16 = 199$ remainder $12 \rightarrow \mathrm{C}$
- $199 \div 16 = 12$ remainder 7

$12 = \mathrm{C}$

So, $3196_{10} = (\mathrm{C7C})_{16}$

Correct Answer: Option B: C7C

Topic : Previous Year Questions + Doubt Class

#Q. Given below are two statements :

Statement I : 011010 is 2's complement representation of −37.

Statement II : (211) 8 is octal representation of 137.

In the light of the above Statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but statement II is false

(D) Statement I is false but statement II is true

Ans. D

Solution:

Statement I: "011010 is the 2's complement representation of -37."

2's Complement Representation of -37:

- Binary of 37 in 6 bits: 100101.
- 1's complement of 100101: 011010.
- Adding 1 to 011010 gives 011011, not 011010.
- 011010 cannot represent -37 in 2's complement.

So, Statement I is false.

Statement II: "(211)s is the octal representation of 137."

Octal to Decimal Conversion:

- $(211)_8$ in decimal: : $2\times8^2+1\times8^1+1\times8^0=128+8+1=137$

So, Statement II is true.

The correct answer is(D): Statement I is false but statement II is true

Topic : Previous Year Questions + Doubt Class

#Q. Suppose we have a 10-bit computer that uses 10-bit floating point computational unit (Float number uses IEEE floating-point arithmetic where a floating point number has 1 sign bit, 5 exponent bits, and 4 fraction bits). The representation for +∞ (plus infinity) is

(A) 0 11111 1111

(B) 1 11111 0000

(C) 0 00000 1111

(D) 0 11111 0000

Ans. D

Solution

- 10-bit Floating Point: $+\infty$ Representation

Format: The 10-bit floating point is structured as follows:

- 1 bit for sign (S)
- 5 bits for exponent (E)
- 4 bits for fraction/mantissa (F)

IEEE Rule for $+\infty$:

In IEEE floating-point format, $+\infty$ is represented as:

- Sign bit (S): 0
- Exponent bits (E): all 1s $\rightarrow$ 11111
- Fraction bits (F): all 0s $\rightarrow$ 0000

Final 10-bit Representation: (4) 0111110000

Topic : Previous Year Questions + Doubt Class

#Q. Given the following truth table:

A	B	x
0	0	1
0	1	0
1	0	0
1	1	0

Which of the following Boolean functions does it represent?

(A) OR
(B) XOR
(C) NOR
(D) XNOR

Ans. C
Solution:
Only true when A = 0, B = 0
This matches the output of NOR = NOT (A OR B).

A	B	A OR B	NOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

This matches exactly with the given truth table.
Correct answer: Option C (NOR)

Topic : Previous Year Questions + Doubt Class

#Q. Which of the following storage hardwares can be used as a back-up device?

A. HDD
B. ROM
C. RAM
D. Cache
E. Magnetic Tape

Choose the correct answer from the options given below:

(A) Only A, B and C
(B) Only B, C and D
(C) Only A and E
(D) Only A, C and E

Ans. C

Solution:

A. HDD (Hard Disk Drive)

- Yes, HDDs are commonly used for backup storage due to their large capacity and cost-effectiveness.

B. ROM (Read-Only Memory)

- ROM is non-volatile, but it's not writable under normal operations, so it cannot be used for backup of user data.

C. RAM (Random Access Memory)

- RAM is volatile memory — contents are lost when power is off. So it’s not suitable for backup.

D. Cache

- Cache is small, volatile, and temporary storage used by the CPU. It cannot be used for backup.

E. Magnetic Tape

- Magnetic tape is a classical backup medium, especially in enterprise systems due to its durability and low cost for archiving.

Correct Devices for Backup:

- A. HDD
- E. Magnetic Tape

Correct Answer: Option C — Only A and E

2 Mins Summary

↳ P1Q



Homework from SAMARTH CUET PG 2027 Module

→ Revision

+ Revise DPP

Yt Replay:

↓

Revision + Q's

THANK YOU